SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL 1

Analytical Problem Solving

COURSE NAME: Deep Learning **COURSE CODE: MLA0403**
FACULTY : Dr . Arul Raj A.M

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

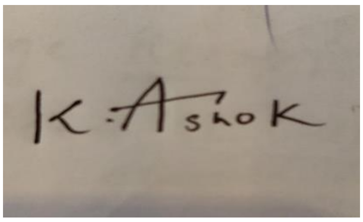
Assessment Tool Weightage: 35%

Duration: 30 minutes

Marks: 25

Code of Conduct:

I K Ashok (192425322) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



K Ashok

Signature of the student: **Name of the student:**

QUESTIONS:05 **Total Marks:25**

Analytical Problem Solving

1. Learning Algorithms (Optimization Behavior)

A machine learning model is trained using gradient descent on a convex loss function. Initially, the learning rate is set very high, and later it is gradually decreased.

Question:

Analyze how the choice of a high initial learning rate followed by decay affects convergence. Under what conditions can this strategy outperform a constant learning rate? Discuss possible risks and justify with reasoning.

Answer :

Gradient descent updates model parameters by moving in the opposite direction of the gradient. The learning rate determines the size of each update. If the learning rate is very high in the beginning and gradually decreases over time, this strategy is called learning rate decay. Initially using a high learning rate allows the optimizer to make large steps. This helps the model reach the region of the global minimum much faster than a very small constant learning rate. Large updates also help the optimizer avoid getting stuck in flat regions and speed up the initial learning process. As training progresses, decreasing the learning rate enables smaller and more precise parameter updates. These fine adjustments help the algorithm converge smoothly toward the optimal solution with less oscillation. This approach can outperform a constant learning rate when the loss function is convex, the initial learning rate is chosen carefully, and the decay schedule is appropriate. Common decay methods include step decay, exponential decay, and cosine annealing.

However, there are risks. If the initial learning rate is too large, the optimizer may overshoot the minimum, causing instability or divergence. If the learning rate decays too quickly, the optimizer may stop learning before reaching the optimal solution. Therefore, a high initial learning rate followed by gradual decay provides both fast learning and accurate convergence, making it one of the most widely used optimization strategies in machine learning.

2. Maximum Likelihood Estimation (MLE)

Suppose you are given a dataset generated from a Gaussian distribution with unknown mean μ and known variance σ^2

Question:

Derive the Maximum Likelihood Estimator for μ, and analytically explain how the estimate changes when:

- The dataset contains outliers
- The sample size increases

  What does this imply about the robustness of MLE?

Answer :

Assume x1, x2, …, xn are independent samples drawn from a Gaussian distribution with unknown mean μ and known variance σ^2. The likelihood function is maximized by differentiating the log-likelihood with respect to μ and setting it equal to zero. The resulting Maximum Likelihood Estimator is:

$\hat{\mu} = (1/n)$

Σxi Hence, the MLE of the mean is simply the sample average.

Effect of Outliers: Since every observation contributes equally to the sample mean, extreme values significantly influence the estimate. Even a single large outlier can shift the estimated mean away from the true population mean. Therefore, the Gaussian MLE is not robust to outliers. Effect of Increasing Sample Size: As the sample size increases, the estimator becomes more accurate. According to the Law of Large Numbers, the sample mean converges to the true population mean. The estimator variance decreases according to $Var(\hat{\mu})=\sigma^2/n$, improving precision. Robustness: MLE is unbiased, efficient, and consistent when the Gaussian assumption is correct. However, it performs poorly when data contains many outliers or does not follow a normal distribution. In such situations, robust estimators such as the median or trimmed mean are preferred.

3. Building a Machine Learning Algorithm (Model Selection)

You are designing a classification algorithm for a dataset with 10,000 features but only 500 training samples.

Question:

Analyze the challenges involved in training such a model. Propose a strategy (algorithm pipeline) to handle this situation and justify each step (e.g., feature selection, regularization, dimensionality reduction). Explain how your approach mitigates overfitting.

Answer :

A dataset with 10,000 features but only 500 training samples suffers from the curse of dimensionality. The model has many more features than examples, increasing the risk of overfitting and computational complexity. Proposed Pipeline: 1. Data Cleaning – Remove missing values and normalize numerical features. 2. Feature Selection – Use correlation analysis, mutual information, chi-square test, or LASSO to remove irrelevant features. 3. Dimensionality Reduction – Apply Principal Component Analysis (PCA) to reduce redundancy while preserving maximum variance. 4. Model Selection – Train a regularized Logistic Regression, Linear SVM, or Random Forest depending on the problem. 5. Hyperparameter Tuning – Use k-fold cross-validation to select the best regularization parameter. 6.

Overfitting Prevention – Apply L1/L2 regularization, early stopping, dropout (for neural networks), and validation-based model selection. 7. Final Evaluation – Evaluate on an unseen test dataset using accuracy, precision, recall, F1-score, and ROC-AUC.

4. Neural Networks (Multilayer Perceptron - MLP)

Consider a Multilayer Perceptron with one hidden layer using sigmoid activation functions.

Question:

Explain analytically why adding more hidden neurons increases the representational power of the network. Then, reason why this does not always lead to better performance. Support your answer using concepts like overfitting and generalization.

Answer :

A Multilayer Perceptron (MLP) with one hidden layer uses nonlinear activation functions such as sigmoid to learn complex relationships. Increasing the number of hidden neurons increases the network's representation power because more neurons can learn more nonlinear features and decision boundaries. According to the Universal Approximation Theorem, a sufficiently large hidden layer can approximate many continuous functions. However, increasing neurons also increases the number of trainable parameters. When the training dataset is small, the network may memorize training samples instead of learning general patterns. This results in overfitting, where training accuracy becomes high but test accuracy decreases. Generalization is the ability of a model to perform well on unseen data. Better generalization requires balancing model complexity and available training data. Regularization methods such as dropout, weight decay (L2 regularization), batch normalization, and early stopping help improve generalization. Therefore, adding hidden neurons increases learning capacity but does not always improve performance. The optimal architecture depends on data size, complexity, and proper regularization.

5. Backpropagation, SGD & Curse of Dimensionality

A deep neural network is trained using Stochastic Gradient Descent (SGD) on a very high-dimensional dataset.

Question:

Analyze how the **curse of dimensionality** impacts:

- Gradient updates in backpropagation
- Convergence speed of SGD
- Model generalization

Propose at least two techniques to overcome these issues and justify them analytically.

Answer :

The curse of dimensionality refers to problems that arise when the number of features becomes very large. Impact on Gradient Updates: In high-dimensional spaces, gradients become noisy and may point in inconsistent directions. This slows learning and makes optimization unstable. Impact on SGD Convergence: Stochastic Gradient Descent requires more iterations because the optimizer searches through a much larger parameter space. The variance of gradient estimates also increases, reducing convergence speed.

Impact on Generalization: High-dimensional models often contain many unnecessary parameters, making them prone to overfitting. As a result, the model performs well on training data but poorly on unseen data. Techniques to Overcome These Problems: 1. Feature Selection and PCA – Reduce the number of input dimensions while retaining important information. 2. Regularization – Apply L1, L2, dropout, or early stopping to reduce overfitting. 3. Adaptive Optimizers – Adam and RMSProp automatically adjust learning rates for faster convergence. 4. Data Augmentation and More Training Data – Increase the diversity of training samples, improving generalization. In conclusion, reducing dimensionality and applying regularization significantly improve optimization, convergence speed, and model performance.